Tune-Up Sequence for
Mechanical Governor

Before starting an engine after an engine speed control adjustment or after removal of the engine governor cover, the service technician must determine that the injector racks move to the no-fuel position when the governor stop lever is placed in the stop position. Engine over speed will result if the injector racks cannot be positioned at no-fuel with the governor stop lever. An over-speeding engine can result in engine damage which could cause personal injury. The technician should also ensure that the racks are capable of full fueling the engine

1. Adjust the exhaust valve clearance cold.
2. Time the fuel injectors.
3. Adjust the governor gap.
4 Position the injector rack control levers
5 Adjust the maximum no load speed.
6. Adjust the idle speed
7 Adjust the Belleville spring for horsepower, (TT or torque tailored engines only).
8 Adjust the buffer screw
9. Adjust the throttle booster spring (variable speed governor only)
10. Adjust the supplementary governing devices if used.

On-highway vehicle and coach engines built by Detroit Diesel Corporation are certified to be in compliance with Federal and California Emission Regulations established for each model year beginning with 1970.
These engines no longer meet the emission standards of today but as long as they meet the standards prevailing in the year they were manufactured they are legal on the road. Due to the propensity of coach fleets such as the TTC and others to overhaul instead of replace we will see these engines in service for some time to come.
Emission certification is dependent on five physical
Characteristics:
1. Fuel Injector type
2. Maximum full load engine speed
3. Camshaft timing
4. Fuel injector timing.
5. Throttle delay (orifice size).

EXHAUST VALVE CLEARANCE ADJUSTMENT
The correct dimension for the exhaust valve clearance and injector timing dimension can be found on the information plate on the rocker cover or in the engine service manual.
The correct exhaust valve clearance at normal engine operating temperature is important for smooth, efficient operation of the engine. Insufficient valve clearance can result in loss of compression, misfiring cylinders and, eventually, burned valve seats and valve seat inserts. Excessive valve clearance will result in noisy operation increased valve face wear and valve lock damage
Whenever the cylinder head is overhauled, the exhaust valves are reconditioned or replaced or the valve operating mechanism is replaced or disturbed in any way, the valve clearance must he adjusted to the cold setting to allow for normal expansion of the engine parts during the engine warm-up period. This will ensure a valve setting that is close enough to the specified clearance to prevent damage to the valves when the engine is started.
All of the exhaust valves can he adjusted in Firing order sequence during one full revolution of the crankshaft. Refer to the manual for the correct engine firing order.
This is the Detroit procedure for adjustment however looking at the adjustment sequence above we see that the next adjustment is the Injector Timing so it makes sense to perform both of these adjustments at the same time as the injector timing adjustment will require the engine to be cranked over as well. We can start at any point and simply by observing the valves and injectors we can determine what is happening in a particular cylinder. If an injector plunger is being depressed by the rocker we know that that cylinders piston is at the top of its stroke and because it is a two stroke we know that the valves will not be actuated until the piston is moving towards the bottom of the stroke, therefore we can adjust the valves on that cylinder. At the same time on a separate cylinder we will observe the exhaust valves being depressed by the rockers. This tells us that this cylinder is close to the bottom of its power stroke which means we can adjust injector height, (timing). By following this procedure carefully we can adjust all of the valves and injectors in one rotation of the engine.
The actual valve adjustment procedure follows;